

FOHBOH | MGE™ · METRICS GOVERNANCE ENGINE

Developer Guide

Certify evidence, read sealed vault records and verify every seal, through MGE's versioned API. Every example is real MGE output, produced from synthetic data.

API	CONTRACT	EDITION	DATE
v1	1.11	Version 1.0	October 10, 2026

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In this guide

The machine-readable reference (OpenAPI 3.1) is served by MGE at `GET /mge/v1/openapi.json`; a dated copy is at `fohboh.ai/fohboh-mge-openapi.json`. Questions: `hello@fohboh.com`.

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Overview

MGE, the Metrics Governance Engine, **certifies evidence**. You send one location's evidence for one module and one fiscal period; MGE runs a deterministic certification, with no AI anywhere in it, and seals the result as an append-only **vault record** carrying a Trust Score (0–100), an outcome, certified findings, reconciliations and a cryptographic seal that anyone can verify.

The API has two doors and a public surface. Each key opens exactly one door.

Door	Who uses it	Endpoints
Write (Contract №1)	Whoever submits evidence	POST /certify · POST /certify/batch · GET /jobs/{id} · GET /modules · GET /modules/{id}/schema
Read (Contract №2)	Whoever displays results	GET /vault · GET /vault/{id} · GET /vault/{id}/evidence · GET /vault/{id}/exportpack · GET /recoveries · GET /rules
Public (no key)	Anyone	GET /vault/{id}/verify · GET /keys · GET /openapi.json

Base URL. Production: <https://api.fohboh.ai/mge/v1>. Your sandbox host and keys arrive with your access sheet during onboarding. The machine-readable reference, OpenAPI 3.1 generated from the same contract package, is always at [GET /mge/v1/openapi.json](#); a dated copy is available to download.

MGE certifies; it does not recover money or give advice. It holds no disputes, letters or tasks. Your product presents MGE's certified facts under your own brand.

Quickstart

- 1. Get sandbox keys.** During onboarding FohBoh issues a write key and a read key (`mge_write_sbx_...`, `mge_read_sbx_...`). Every sandbox certification is a TEST RUN: never billed, deletable, and never written to write-once storage.
- 2. Submit evidence** for one location, one module, one period:

```
curl -X POST https://api.fohboh.ai/mge/v1/certify \
  -H "Authorization: Bearer $MGE_WRITE_KEY" \
  -H "Content-Type: application/json" \
  -H "Idempotency-Key: loc-0142:M02:doordash:2026-08-01:r1" \
  --data @m02-august.json
```

```
{
  "module_id": "M02",
  "location_id": "loc-0142",
  "source_system": "doordash",
```

```

"period": {
  "start": "2026-08-01",
  "end": "2026-08-31"
},
"idempotency_key": "loc-0142:M02:doordash:2026-08-01:r1",
"records": [
  {
    "record_id": "AGR-1",
    "record_type": "dsp_agreement",
    "dsp": "doordash",
    "agreement_id": "DD-MA-2026",
    "agreement_sha256": "436bfe7866cfe9535a210904c9c6ac09a51780350fa7561e1ae28bb6c0f45144",
    "signed_date": "2025-12-15",
    "effective_start": "2026-01-01",
    "plan_tier": "Plus",
    "commission_bps": 2000,
    "commission_basis": "subtotal_ex_tax"
  },
  {
    "record_id": "FT-ERR",
    "record_type": "dsp_fee_term",
    "dsp": "doordash",
    "agreement_id": "DD-MA-2026",
    "charge_class": "error_charge",
    "authorized": true,
    "per": "order",
    "max_per_order": 1
  },
  {
    "record_id": "STMT-08",
    "record_type": "dsp_statement",
    "dsp": "doordash",
    "statement_id": "DD-2026-08",
    "period_start": "2026-08-01",
    "period_end": "2026-08-31",
    "issued_date": "2026-09-03",
    "source_format": "doordash.merchant_statement.v1",
    "order_count": 812,
    "gross_sales_cents": 3150000,
    "net_payout_cents": 2517850,
    "source_document_sha256": "1fa6d2048f8ab60e2d4a3154d46c66337ba7ee4decaac6654c908ac4228c22ff"
  },
  "... 6 more records: statement lines, an order charge, POS sales, a bank deposit, fulfillment proof
  ...
]
}

```

```

{
  "job_id": "job_576765b2e8185a33c4082ace",
  "status": "queued",
  "record_count": 9,
  "estimated_ms": 200,
  "accepted_at": "2026-09-24T15:02:17.000Z"
}

```

3. The body (abridged; the complete synthetic example is in Submit evidence): MGE accepts the work and answers at once with `202 Accepted`:

4. Poll the job until `status` is `certified` or `failed`:

```
curl https://api.fohboh.ai/mge/v1/jobs/job_576765b2e8185a33c4082ace \
-H "Authorization: Bearer $MGE_WRITE_KEY"
```

```
{
  "job_id": "job_576765b2e8185a33c4082ace",
  "status": "certified",
  "module_id": "M02",
  "location_id": "loc-0142",
  "vault_id": "vr_a2e1f8d6429ae4afb0f912833c8c6827",
  "error": null,
  "created_at": "2026-10-10T01:10:33.330Z",
  "record": "... the full vault record (below) ..."
```

5. Read the sealed record with your read key:

```
curl https://api.fohboh.ai/mge/v1/vault/vr_a2e1f8d6429ae4afb0f912833c8c6827 \
-H "Authorization: Bearer $MGE_READ_KEY"
```

6. The outcome is in `certified_metrics.outcome`; a certified job is not the same as a CERTIFIED outcome (see Outcomes).

7. Verify the seal, no key needed, so your client or a counterparty can check it too:

```
curl "https://api.fohboh.ai/mge/v1/vault/vr_a2e1f8d6429ae4afb0f912833c8c6827/verify?
sha256_hash=a2e1f8d6429ae4afb0f912833c8c6827d378e5dc711dd126067191e2700f8b54"
```

```
{
  "vault_id": "vr_a2e1f8d6429ae4afb0f912833c8c6827",
  "valid": true,
  "superseded_by": null,
  "signature_valid": true
}
```

03 SECURITY

Authentication & keys

Every call except the public ones sends a bearer key:

```
Authorization: Bearer mge_write_sbx_<your key>
```

Each key is scoped to **integration × location group × door**, and to one **environment**:

Scope	Meaning
Integration	Your integration id, for example acme-pos.
Location group	A set of your client's locations. A key acts only on locations inside its group.
Door	write or read. A write key cannot read vault records; a read key cannot submit evidence. GET /jobs/{id} and GET /modules accept either.
Environment	sandbox (mge_..._sbx...): every certification is a TEST RUN. production (mge_write..., mge_read...): real, billable certifications, issued only after FohBoh's production approval.

- **Shown once.** FohBoh shows a key one time when it is issued; MGE keeps only its SHA-256 fingerprint. Store it in your secret manager immediately. FohBoh never sends keys by email.
- **Expiry.** Sandbox keys last 30 days and production keys up to a year. Every authenticated response carries an `MGE-Key-Expires` header, so you can rotate in good time.
- **Rotation.** Ask FohBoh for a new key, switch over, then have the old key revoked. Both work during the overlap.
- **No OAuth.** MGE uses scoped API keys only; there is no user sign-in on the API.

Situation	Status	error.code
No Authorization header, or an unknown or revoked key	401	UNAUTHENTICATED
An expired key (rotate before the date in <code>MGE-Key-Expires</code>)	401	KEY_EXPIRED
The key's door cannot use this endpoint (for example a read key on /certify)	401	KEY_SCOPE
The location is outside the key's location group	401	LOCATION_NOT_IN_SCOPE
A record that belongs to another account	404	NOT_FOUND (MGE never confirms that it exists)

A read key used on `/certify`, as returned by MGE:

```
{
  "error": {
    "code": "KEY_SCOPE",
    "message": "A read key cannot use this endpoint"
  },
  "request_id": "req_433gh1x8sf"
}
```

04 REFERENCE

Endpoints

Method and path	Key	Purpose
-----------------	-----	---------

POST /certify	write	Submit one certification (202, asynchronous).
POST /certify/batch	write	Submit 1–50 certifications in one call.
GET /jobs/{job_id}	write or read	Job status; the full vault record once certified.
GET /modules	write or read	The live module registry: status and maximum records per request.
GET /modules/{module_id}/schema	write or read	The JSON Schema of records[] for that module.
GET /vault	read	Your sealed records, newest first, with filters.
GET /vault/{vault_id}	read	One sealed record.
GET /vault/{vault_id}/evidence	read	The evidence index, or a Merkle inclusion proof with ? record_id=.
GET /vault/{vault_id}/exportpack	read	The ExportPack: received data, evidence with inclusion proofs, the record, narrative, keys and a SHA-256 manifest.
GET /recoveries	read	The recovery ledger: which credits paid back earlier findings.
GET /rules	read	Rule counts by category. Rule identifiers, logic and weights are never exposed.
GET /actions · POST /actions/{action_id}/resolve · GET /patterns	read	Follow-ups MGE raises after certification, and weekly patterns across records, for integrations that act on them.
GET /vault/{vault_id}/verify	none	Public seal verification.
GET /keys	none	MGE's public signing keys, including retired ones.
GET /openapi.json	none	OpenAPI 3.1 reference.

All paths are relative to the base URL ending in `/mge/v1`.

05 CONTRACT №1

Submit evidence: POST /certify

One request certifies **one location × one module × one period**, with up to **10,000 records**.

Field	Type	Required	Notes
module_id	M01 · M02 · M04	yes	Only active modules certify (see Modules).
location_id	string	yes	Must be inside your key's location group.
source_system	string	yes	A canonical key for the module, for example doordash.

period	{ start, end }, ISO dates	yes	Exactly one fiscal period of your client's calendar (a calendar month by default). M01 uses one processor statement period of 1–35 days.
records	array, 1–10,000	yes	The module's evidence records; each has a record_id unique in the request and a record_type.
idempotency_key	string, 1–128	no	Or the Idempotency-Key header. See Idempotency.
test_run	boolean	no	A TEST RUN is certified twice and never supersedes. Always true on sandbox keys.
direction	UNILATERAL · INBOUND · OUTBOUND	no	Default UNILATERAL. INBOUND and OUTBOUND need a bidirectional plan.
paired_vault_id	vr_...	OUTBOUND only	The INBOUND certification this decision answers.

The complete synthetic M02 (delivery fees) request used throughout this guide. It validates against the contract and certified as shown:

```
{
  "module_id": "M02",
  "location_id": "loc-0142",
  "source_system": "doordash",
  "period": {
    "start": "2026-08-01",
    "end": "2026-08-31"
  },
  "idempotency_key": "loc-0142:M02:doordash:2026-08-01:r1",
  "records": [
    {
      "record_id": "AGR-1",
      "record_type": "dsp_agreement",
      "dsp": "doordash",
      "agreement_id": "DD-MA-2026",
      "agreement_sha256": "436bfe7866cfe9535a210904c9c6ac09a51780350fa7561e1ae28bb6c0f45144",
      "signed_date": "2025-12-15",
      "effective_start": "2026-01-01",
      "plan_tier": "Plus",
      "commission_bps": 2000,
      "commission_basis": "subtotal_ex_tax"
    },
    {
      "record_id": "FT-ERR",
      "record_type": "dsp_fee_term",
      "dsp": "doordash",
      "agreement_id": "DD-MA-2026",
      "charge_class": "error_charge",
      "authorized": true,
      "per": "order",
      "max_per_order": 1
    },
    {
      "record_id": "STMT-08",
```

```

    "record_type": "dsp_statement",
    "dsp": "doordash",
    "statement_id": "DD-2026-08",
    "period_start": "2026-08-01",
    "period_end": "2026-08-31",
    "issued_date": "2026-09-03",
    "source_format": "doordash.merchant_statement.v1",
    "order_count": 812,
    "gross_sales_cents": 3150000,
    "net_payout_cents": 2517850,
    "source_document_sha256": "1fa6d2048f8ab60e2d4a3154d46c66337ba7ee4decaac6654c908ac4228c22ff"
  },
  {
    "record_id": "STMT-08-L1",
    "record_type": "dsp_statement_charge",
    "dsp": "doordash",
    "statement_id": "DD-2026-08",
    "line_number": 1,
    "charge_class": "commission",
    "description": "Commission",
    "amount_cents": 630000,
    "basis_cents": 3150000,
    "rate_bps": 2000
  },
  {
    "record_id": "STMT-08-L2",
    "record_type": "dsp_statement_charge",
    "dsp": "doordash",
    "statement_id": "DD-2026-08",
    "line_number": 2,
    "charge_class": "error_charge",
    "description": "Error charges",
    "amount_cents": 2150
  },
  {
    "record_id": "OC-50117-1",
    "record_type": "dsp_order_charge",
    "dsp": "doordash",
    "order_id": "ORD-50117",
    "order_date": "2026-08-14",
    "statement_id": "DD-2026-08",
    "line_number": 1,
    "charge_class": "error_charge",
    "amount_cents": 2150
  },
  {
    "record_id": "POS-08",
    "record_type": "pos_delivery_sales",
    "dsp": "doordash",
    "period_start": "2026-08-01",
    "period_end": "2026-08-31",
    "order_count": 812,
    "gross_sales_cents": 3150000,
    "source_document_sha256": "61a4a548153e305115e658ba2ab26199c9e2e1a92cec17106674378bbced08cd"
  },
  {
    "record_id": "DEP-0904",
    "record_type": "bank_deposit",
    "dsp": "doordash",

```

```

    "deposit_date": "2026-09-04",
    "amount_cents": 2517850,
    "source_document_sha256": "1ec72718b303e96406a5a58da363de2df5d0c3d904cca1addddcc0779e6875e9"
  },
  {
    "record_id": "PF-50117",
    "record_type": "order_fulfillment",
    "dsp": "doordash",
    "order_id": "ORD-50117",
    "proof_type": "kds_completion",
    "captured_at": "2026-08-14T19:42:10-05:00",
    "document_sha256": "373d279185f44d9cababbfe3678dbaabc433e38dee2131f8b711eb6c55925def"
  }
]
}

```

Batch. `POST /certify/batch` takes `{ "requests": [ ... 1-50 ... ] }` and returns `202 { "jobs": [ ... ] }` in request order. A batch is **not atomic**: MGE stops at the first request it refuses and returns that error; the requests before it were accepted. Put an `idempotency_key` on every batch request so resending after a fix is safe.

06 ASYNCHRONOUS

Jobs & polling

status	Meaning
queued	Accepted and stored; accepted work survives a server restart.
processing	Being certified.
certified	Sealed. vault_id and the full record are present.
failed	Not sealed; error.code says why. No vault record exists.

Poll with backoff: start near `estimated_ms`, then slow down. Webhooks are not available yet.

Job failure code	Meaning	What to do
PERIOD_NOT_FISCAL	The period is not one whole fiscal period (M01: not a 1–35 day statement period).	Send the fiscal or statement period.
SCHEMA_VIOLATION	A record failed an intake check the request schema cannot express.	Fix the record and resubmit.
NO_STATEMENT · NO_INVOICE	The module's anchor document is missing.	Include the statement or invoice.
ORPHAN_LINE	A line with no statement or invoice header.	Include the header.
DUPLICATE_ORDER · DUPLICATE_STATEMENT	An id appears twice.	De-duplicate.
MIXED_PROCESSOR	M01: records from more than one processor.	Certify each processor account separately.

MODULE_NOT_ACTIVE	The module stopped accepting certifications.	Contact FohBoh.
DETERMINISM_FAILURE · INTERNAL_ERROR	An MGE-side integrity or processing error, not your data.	Retry with the same idempotency key, then contact FohBoh.

07 DETERMINISTIC BY DESIGN

Request conventions

- **Money is integer cents**; rates are integer basis points (`2000` = 20.00%). A float in a money field is refused with `400 SCHEMA_VIOLATION`; MGE never rounds for you.
- **Quantities and weights** are decimal strings, for example `"12.5"`, at most 4 decimal places.
- **Dates** are `YYYY-MM-DD`; timestamps are ISO 8601 with an offset.
- **Document hashes** are lowercase 64-hex SHA-256 of the source document you hold.
- `record_id` is your own id (1-128 characters), unique in the request; findings, proofs and gaps cite it back to you. Record order does not matter.

08 COVERAGE

Modules

Module	What it certifies	source_system keys	Status
M01 Merchant Fee Recovery	Card processor charges against the processing agreement	toast heartland square stripe worldpay fiserv chase other_processor	Active
M02 Delivery Fee Recovery	Delivery platform charges against the merchant agreement	doordash ubereats grubhub	Active
M04 Supply Chain Certification	Distributor invoices against contract prices and receiving	sysco usfoods pfg gfs other_distributor	Active
M03 Royalty Fee Recovery	Franchisor royalty and fund charges	franchisor_statement pos_export	Not yet active

Fetch each module's exact record schema from `GET /modules/{module_id}/schema`; it is generated from the contract and is always current. Field-by-field record guides for each module are shared during onboarding.

09 CONTRACT № 2

Response schema: the vault record

Field	Meaning
-------	---------

vault_id	Record id (vr_ + 32 hex), derived from the seal hash.
cert_ref	Human reference, for example CRT-2026-0001.
tenant_id · location_id · module_id · period	As certified.
fiscal_period · period_label · cadence	The client's fiscal period and cadence.
state	S1-S7 (see Outcomes).
certified_at	An input to the seal.
sha256_hash	The seal hash: use it for public verification.
content_sha256	The content fingerprint: identical for any two certifications of the same evidence under the same settings and versions.
merkle_root	Root over your submitted evidence records (RFC 6962).
formula_version · rule_set_version · sector_pack	The pinned versions the record was certified under.
certified_metrics	trust_score, outcome, hard_gate_overrides, penalty_rows (where coverage was lost and how to fix it), disputed_total_cents, findings, kpis, reconciliation, and where present recoveries, evidence_gaps, held_items.
narrative	The sealed plain-language certification narrative and its SHA-256 (DCLS, the Deterministic Certified Language System).
manifest_sha256 · stage_chain	Hashes of the certification manifest and of each Trust Loop stage.
seal	Custody metadata: standard, custodian, sealed time, jurisdiction, legal hold.
signature	MGE's signature over the sealed record (ED25519 or ECDSA_P256_SHA256); match key_id in GET /keys.
evidence_bundle	Receipt for the write-once evidence bundle; null for TEST RUNS.
superseded_by · test_run	Set when a later certification replaced this one; whether it is a TEST RUN.

Findings carry `ref` (`F-001` for recoveries, `V-001` for items needing review), `kind`, a plain-language `label` and `description`, `amount_cents`, and the `evidence` record ids that support it. Build your interface from these fields; MGE never exposes rule identifiers, rule logic or scoring weights.

The record MGE returned for the example (abridged where marked):

```
{
  "vault_id": "vr_a2e1f8d6429ae4afb0f912833c8c6827",
  "cert_ref": "CRT-2026-0001",
  "tenant_id": "acme-pos-demo",
  "location_id": "loc-0142",
  "module_id": "M02",
  "period": {
```

```

    "start": "2026-08-01",
    "end": "2026-08-31"
  },
  "fiscal_period": "2026-08",
  "cadence": "fiscal_period",
  "period_label": "2026-08",
  "state": "S1",
  "certified_at": "2026-09-24T15:02:18.000Z",
  "sha256_hash": "a2e1f8d6429ae4afb0f912833c8c6827d378e5dc711dd126067191e2700f8b54",
  "content_sha256": "3f395fd3136918230a023d827410dcbbd6571f057fa6b0131ead39a7b48ec16f",
  "merkle_root": "dafc82c84e961b3b95498bfeeb5794437f21cbc594cd9258299202a063354fcc",
  "formula_version": "m02_dfr_v3",
  "rule_set_version": "DCLS-2.0-LEGAL@a51868bf6a2cbe14",
  "sector_pack": {
    "id": "INST-01",
    "name": "Restaurant MGE (INST-01 Restaurant pack)",
    "version": "1.2.0"
  },
  "certified_metrics": {
    "trust_score": 100,
    "outcome": "CERTIFIED",
    "hard_gate_overrides": [],
    "penalty_rows": [],
    "disputed_total_cents": 2150,
    "findings": [
      {
        "ref": "F-001",
        "kind": "RECOVERY",
        "label": "Chargeback or error charge not supported by the evidence",
        "description": "Order ORD-50117 (2026-08-14) was charged an error charge of $21.50; the restaurant's kitchen display completion show the order was fulfilled",
        "amount_cents": 2150,
        "evidence": [
          "OC-50117-1",
          "PF-50117",
          "STMT-08"
        ]
      }
    ]
  },
  "kpis": {
    "bank_deposits_cents": 2517850,
    "commission_charged_cents": 630000,
    "commission_contracted_cents": 630000,
    "credits_cents": 0,
    "dsp_gross_sales_cents": 3150000,
    "error_charges_cents": 2150,
    "marketing_charged_cents": 0,
    "net_payout_cents": 2517850,
    "order_count": 812,
    "pos_gross_sales_cents": 3150000,
    "pos_order_count": 812,
    "statement_count": 1,
    "total_charges_cents": 632150
  },
  "reconciliation": [
    {
      "leg": "DD-2026-08: gross sales POS↔DSP",
      "left_cents": 3150000,
      "right_cents": 3150000,

```

```

        "variance_cents": 0,
        "status": "MATCHED"
    },
    {
        "leg": "DD-2026-08: statement footing",
        "left_cents": 2517850,
        "right_cents": 2517850,
        "variance_cents": 0,
        "status": "MATCHED"
    },
    "... every reconciliation leg ..."
]
},
"seal": {
    "fre_standard": "FRE 902(13)/(14) – self-authenticating electronic records",
    "custodian": "FohBoh MGE",
    "sealed_at": "2026-09-24T15:02:18.000Z",
    "jurisdiction": "US",
    "legal_hold": false
},
"narrative": {
    "text": "DCLS certification narrative. Module M02 · location loc-0142 · fiscal period 2026-08 (2026-08-01 to 2026-08-31). 9 evidence records are committed under Merkle root
dafc82c84e961b3b95498bfeeb5794437f21cbc594cd9258299202a063354fcc.\n1 certifie ...",
    "sha256": "be7362e0f3dbd0ad8891358a14555abdd5245a34da6403b2bf168a29e194f3a"
},
"manifest_sha256": "bba894b78481942bc7b2cfd9095140359d26c5475676e359f5a27c505d2e6f73",
"stage_chain": [
    {
        "stage": "RESOLVE",
        "output_sha256": "3447a5f847cf90e9f70250aa64eac381b4f4ae33f373fed201769d59f133ceda",
        "link_sha256": "3703948c3767c0fd40381cdab5ce45bfb72a4141abcd109cba7b251d08052ec"
    },
    {
        "stage": "GOVERN",
        "output_sha256": "0765eedec4a303e53505f0be9dc3b3a83f55b28baa2b8d13749c84bc6043492d",
        "link_sha256": "c3b50daab9399bbc2af54e7d2c76697b5d6c71b4c752addca28fde26a6bda268"
    },
    "... one entry per Trust Loop stage ..."
],
"signature": {
    "key_id": "mge-seal-docs-1",
    "algorithm": "ED25519",
    "value": "BWVZHLFQDxCNuyFjLndcuycH..."
},
"evidence_bundle": null,
"superseded_by": null,
"test_run": true
}

```

10 PUBLISHED BANDS

Outcomes & states

Outcome	Trust Score	State	Meaning
---------	-------------	-------	---------

CERTIFIED	85–100	S1	Fully certified.
VERIFIED	70–84	S2	Certified with discrepancies.
REVIEWED	60–69, or any score with a hard-gate override	S4	Certified on bounded scope: the available evidence only.
FAILED	0–59	S6	The evidence cannot support certification; gaps are itemized.
—	—	S5	Pending: no sealed record yet.
—	—	S7	Superseded by a later certification.

- **The Trust Score measures evidence coverage, not the number of problems.** Findings never lower it: a record can be CERTIFIED and carry large findings. Missing mandatory evidence overrides the outcome to REVIEWED; it never deducts points.
- **Every outcome is sealed.** A FAILED outcome is a sealed record; a failed job produced no record.
- **Sealed records are never updated or deleted.** To correct evidence, submit a new certification for the same location, module, period and direction; the earlier record moves to S7 and names its replacement in `superseded_by`. TEST RUNs never supersede.

11 TRUST, BUT VERIFY

Verification

Public check, no key: `GET /vault/{vault_id}/verify?sha256_hash=<64 hex>` returns `valid`, `superseded_by` and `signature_valid`, and reveals nothing else about the record.

```
{
  "vault_id": "vr_a2e1f8d6429ae4afb0f912833c8c6827",
  "valid": true,
  "superseded_by": null,
  "signature_valid": true
}
```

Prove one evidence record was included without revealing the others: `GET /vault/{vault_id}/evidence?record_id=...` returns an RFC 6962 inclusion proof. Leaf hash = SHA-256(0x00 || leaf); node hash = SHA-256(0x01 || left || right); the result must equal `merkle_root`.

```
{
  "vault_id": "vr_a2e1f8d6429ae4afb0f912833c8c6827",
  "merkle_root": "dafc82c84e961b3b95498bfeeb5794437f21cbc594cd9258299202a063354fcc",
  "proof": {
    "record_id": "PF-50117",
    "leaf": "{\\"captured_at\\":\\"2026-08-14T19:42:10-05:00\\",\\"document_sha256\\":\\"373d279185f44d9cababbfe3678dbaabc433e38dee2131f8b711eb6c55925def\\",\\"dsp\\":\\"doordash\\",\\"order_id\\":\\"ORD-50117\\",\\"proof_type\\":\\"kds_completion\\",\\"record_id\\":\\"PF-50117\\",\\"record_type\\":\\"order_fulfillment\\"}",
    "index": 4,
    "tree_size": 9,
    "proof": [
      "e82e7108c7a1baaa247f60f69f447c5c32ce736906fb13a14ba76b8a9d47dc49",
      "6eb0a176ed0e594e680a10f151447abd208fea93d3d1e831628adbbd4228464f",
      ""
    ]
  }
}
```

Determinism. Certification is a pure function of the evidence, the client's pinned settings and the pinned formula and rule versions. The same evidence under the same versions gives the same `content_sha256`; the seal hash differs only because `certified_at` is sealed with it. Every TEST RUN is certified twice and must reproduce its own hash.

12 WHEN THINGS GO WRONG

Error handling

Every error, on every endpoint, has the same envelope. Quote `request_id` when you contact FohBoh.

```
{
  "error": {
    "code": "SCHEMA_VIOLATION",
    "message": "Request failed schema validation",
    "violations": [
      {
        "field": "records.0.amount_cents",
        "record_id": "STMT-08-L1",
        "reason": "Expected integer, received float"
      }
    ]
  },
  "request_id": "req_zmpdu2wy8i"
}
```

Status	error.code	When
400	MALFORMED_REQUEST	The body is not valid JSON.
400	SCHEMA_VIOLATION	The request or a record fails the contract schema; violations gives the field path and your record_id.

401	UNAUTHENTICATED · KEY_EXPIRED · KEY_SCOPE · LOCATION_NOT_IN_SCOPE	See Authentication.
403	PRODUCT_NOT_SUBSCRIBED	Your integration's product is not in the account's package.
404	NOT_FOUND	No such job, record, module or evidence record in your account.
409	IDEMPOTENCY_CONFLICT	The key was already used for a different request.
422	MODULE_NOT_ACTIVE · DIRECTION_NOT_CONTRACTED · PAIRING_INVALID	Not certifiable as asked.
429	RATE_LIMITED	Too many requests; wait for the Retry-After seconds.
500	SEAL_BROKEN · INTERNAL_ERROR	An MGE-side problem. Contact FohBoh with the request id.

13 FAIR USE

Rate limits

Limit	Per minute
All requests, per key	1,200
POST /certify and /certify/batch, per key	300
Requests without a key (public verification), per address	120

Over a limit, MGE answers `429 RATE_LIMITED` with a `Retry-After` header in seconds. An address-level firewall limit also applies in front of the API. Batch where you can, poll with backoff, and contact FohBoh if your volume needs higher limits.

14 SAFE RETRIES

Idempotency & retries

- MGE identifies a submission by its `idempotency_key`, then the `Idempotency-Key` header, then the canonical SHA-256 of the request itself, so an identical body resent without a key returns the same job.
- Same key, same request:** `202` with the same `job_id` and its current status. Nothing is re-run and nothing is billed.
- Same key, different request:** `409 IDEMPOTENCY_CONFLICT`.
- Recommended key: `<location_id>:<module_id>:<period.start>:<revision>`. Bump the revision when you deliberately re-certify corrected evidence.
- Retry `429` after `Retry-After`, and `5xx` with backoff and the same key. Never retry a `4xx` unchanged.

Versioning & deprecation

- The major version is in the path (`/mge/v1`). Within a major version, changes are additive only: new optional fields, new endpoints, new values. Ignore fields you do not recognize.
- A breaking change ships as a new major version, announced in advance, with a **90-day transition** during which both versions run.
- The contract is published as versioned schemas generated from one source; `GET /openapi.json` always describes the version you are calling. Partners receive the contract changelog during onboarding.
- Sealed records keep the formula and rule-set versions they were certified under, forever.

Not yet available

Capability	Status
Webhooks	Not built. <code>webhook_url</code> is accepted but not called. Poll <code>GET /jobs/{id}</code> .
M03 Royalty Fee Recovery	Contracted, not active: 422 MODULE_NOT_ACTIVE.
Legal seal level	<code>seal_level</code> : "legal" is accepted; no legal threshold is defined yet, so it produces a standard record.
Job status partial	Reserved in the contract; not produced.

We would rather tell you than have you find out.

Support & terms

- **Onboarding:** partners follow a ten-step track from sandbox to production; production keys are issued only after FohBoh's approval.
- **Commercials:** a license fee plus a per-certification toll; sandbox TEST RUNs, rejected requests and idempotent retries are never billed. See pricing.
- **Terms:** use of the API is subject to your partner agreement and the API terms of use.
- **No advisory services:** the API returns governance and certification outputs only, never financial, accounting, legal, tax, labor or operational advice.
- **Contact:** hello@fohboh.com. Quote the `request_id` from any error.